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era5-to-ccpp-scm-tool: single-column model cases from ERA5 reanalysis for any location on Earth

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Software

- Review
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Summary

Single column models (SCMs) simulate the physics of an atmospheric model, including radiation, turbulence, convection, cloud microphysics, and the land surface, for a single vertical column, with the effects of the resolved dynamics supplied as prescribed forcing rather than computed by the model itself (Randall et al., 1996). Because single column models are computationally frugal, they are widely used for developing and testing parameterizations, and for examining what a physics suite does at a particular place and time. Running an SCM requires a case, which consists of initial profiles of the atmospheric state, large-scale advective tendencies of heat and moisture, vertical motion, geostrophic winds, and land-surface properties, all assembled in the layout that the target model reads. Building such a case has traditionally been an ad hoc, per-study effort, and as a result the case libraries that the community shares consist of a few dozen curated field campaign cases.

`era5-to-ccpp-scm-tool` is a Python package and command line tool that builds SCM cases for the Common Community Physics Package Single Column Model (CCPP-SCM) (*Common Community Physics Package Single Column Model (CCPP-SCM) User and Technical Guide, V7.0.0*, 2024; Heinzeller et al., 2023) from ERA5 reanalysis (Hersbach et al., 2020) for any location on the globe. Given a latitude, longitude, and date range, it extracts the reanalysis fields that are needed, derives the full set of large-scale forcing terms, and writes a case in the DEPHY format (GdR DEPHY, 2023) along with its configuration namelist, so that the case can be run directly in the CCPP-SCM. A complete case is generated with a single command:

```
era5-scm-tool run_full_pipeline \
  --start_date 2019-01-01 --end_date 2019-01-02 \
  --lat 31.7438 --lon -110.0522 \
  --case_name fluxnet_US-Whs --output_dir .
```

We have designed the package to fit into the broader scientific Python ecosystem, building on popular packages such as xarray (Hoyer & Hamman, 2017), NumPy (Harris et al., 2020), and MetPy (May et al., 2022), and it provides both a command line tool as well as API access as a Python module.

Statement of need

The development of this tool was motivated by gaps in how the driving data are obtained and in how the forcing terms are derived. ERA5 is normally accessed through the Copernicus Climate Data Store, which requires an account, an API key, and a request queue whose latency is neither bounded nor reproducible. This is a poor fit for batch work and an awkward dependency for a published workflow. We instead read the NSF NCAR ERA5 archive (European Centre

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for Medium-Range Weather Forecasts, 2019), which is public, requires no authentication, and is published in two places with an identical directory layout: an anonymous S3 bucket that can be read from anywhere, and a filesystem mount on NSF NCAR machines. We verified that the two backends produce bit-identical extractions, so a workflow developed on a laptop reproduces exactly on a high-performance computing system. Forcing derivations such as this are usually implemented as ad hoc solutions based on similar sets of equations, and our goal was to formalize the process by developing this package.

We chose CCPP-SCM as the target model because of its lineage. CCPP-SCM is the single-column driver for the Common Community Physics Package (Heinzeller et al., 2023), and the physics suites that it runs are the same source code, invoked through the same framework, as the physics in NOAA's operational Unified Forecast System. Suites are selected at run time from suite definition files, which makes it inexpensive to push a single case through several physics configurations and to attribute the differences to the physics rather than to the setup (Groot et al., 2026). CCPP-SCM also couples a full land surface model (Noah-MP, Niu et al., 2011), which is a requirement for the land-atmosphere coupling problems that motivated this work, and it reads DEPHY (GdR DEPHY, 2023), a community format, so the cases that this tool writes are not restricted to a single model.

Existing tools for building single-column cases do not fill this gap. SCAM6 (Gettelman et al., 2019) is distributed with the CESM physics that it is built around, and the DEPHY case library distributes curated field campaign cases such as GABLS3 (Bosveld et al., 2014). Both of these constrain a study to the locations and periods that someone has already prepared, while the application that motivated this work, evaluating land-atmosphere coupling against flux tower observations (Novick et al., 2018; Pastorello et al., 2020), requires cases at arbitrary sites generated in bulk. CCPP-SCM itself ships a case generator, `UFS_case_gen.py`, which builds DEPHY cases from Unified Forecast System initial condition and history files on the native cubed-sphere grid (*Common Community Physics Package Single Column Model (CCPP-SCM) User and Technical Guide, V7.0.0*, 2024). That script requires the user to already hold a matched set of initial conditions, supergrid files, and history output, and it is the appropriate choice when the question concerns the forecast system itself. It draws on a forecast archive, however, which is produced by an operational configuration that changes with each upgrade, so a record assembled across several years is not homogeneous. ERA5 is a reanalysis, in which a single frozen model and assimilation system is applied across the whole 1940 to present record while ingesting the observational record as it goes (Hersbach et al., 2020). For studies that span many sites and many years, that observational constraint and temporal homogeneity are of interest, making the two tools complementary rather than competing.

Methods

ERA5 and a DEPHY case file differ in many conventions, and the conversions between them fall into six broad groups.

The first group concerns coordinates and indexing. Horizontal gradients are taken across a stencil, by which we mean the small block of neighboring ERA5 grid cells, three by three by default, that is centered on the site. ERA5 latitudes are stored north first, so a northward gradient is computed as the difference between the first and third rows of the stencil rather than the reverse. Longitudes run from 0 to 360 degrees and are unwrapped when a site lies near the prime meridian, and pressure levels arrive in hPa and are re-indexed onto their converted values rather than simply rescaled.

The second group concerns time. The accumulated radiative fluxes are archived as hourly totals in J/m^2 and are divided by the accumulation period to give rates in W/m^2. Those fluxes come from a forecast stream dimensioned by forecast initial time and forecast hour, which is flattened onto valid time to give an hourly series. The absolute timestamps that ERA5 carries are then converted to seconds since the start of the case, which is the convention

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DEPHY uses.

The third group is thermodynamic. Temperature and specific humidity are converted to liquid water potential temperature and total water specific humidity, the pressure velocity that ERA5 archives is converted to a vertical velocity in m/s using the thermodynamic relations provided by MetPy, and geopotential is converted to geopotential height.

The fourth group is dynamical. Geostrophic winds are computed from the gradients of geopotential height across the stencil, and the horizontal and vertical advective tendencies of liquid water potential temperature and total water are computed using three point derivatives on a non-uniform stencil with spherical metric factors applied. A column mean radiative heating rate is diagnosed from the difference between the net radiative fluxes at the top of the atmosphere and at the surface.

The fifth group concerns the land surface. ERA5 is produced with the HTESSEL land surface model (Balsamo et al., 2009), whose state cannot be handed to Noah or Noah-MP unchanged, and reconciling the two requires three separate transformations. The four soil layers that HTESSEL carries have interfaces at 7, 28, 100 and 289 cm against the historical default four layer configuration at 10, 40, 100 and 200 cm that Noah and Noah-MP use, so soil temperature and moisture are remapped by overlap weighting, which conserves the depth integral of the remapped quantity. The two models also assign different porosities and wilting points to the same soil, so carrying a volumetric water content across unchanged can place it outside the physical range of the receiving soil type, and we therefore transfer the degree of saturation between the wilting point and the porosity of each parameter set rather than the water content itself. This preserves the ratio from which Noah computes its water stress factor, and with it the evaporative regime. The liquid water fraction that Noah carries separately is recovered from the freezing point depression implied by the Clapp-Hornberger retention curve of the receiving soil. Land cover and soil texture are mapped from the HTESSEL classes onto the IGBP and STATSGO classifications that the GFS physics is configured to read, and the green vegetation fraction is obtained by closing the ERA5 cover fractions with the leaf area index, which supplies the seasonal cycle that the cover fractions themselves lack.

The final group concerns the schema. ERA5 short names are mapped onto DEPHY variable names, and the grouped arrays that older CCPP-SCM releases used are written into the flat layout that DEPHY specifies. This conversion is not purely cosmetic, because DEPHY encodes which forcing terms are active in the global attributes of the case file. The attributes are set to enable advection of liquid water potential temperature and total water, to supply pressure velocity and geostrophic forcing, to enable the internal radiation scheme, and to declare whatever nudging the user has asked for. A case whose attributes disagree with the arrays it carries will run to completion and produce a plausible but incorrect answer, which is one of the reasons we consider it worthwhile to package these steps together. The companion Fortran namelist that declares the case to the SCM is written at the same stage.

Configuring a case

A column driven by prescribed forcing has nothing tying it to the reanalysis it was derived from, so the errors that any parameterization makes accumulate without check. Over a day or two this is unimportant, and it is the configuration in which most single column cases are used. Over weeks the column drifts away from the reanalysis and can leave the range in which the physics is numerically well behaved. This package therefore offers two controls, both off by default so that a short case is unaffected.

The first relaxes the column toward the reanalysis profiles, which the DEPHY format supports directly and which the package writes from the same fields it already derives. Three settings are available. Relaxing nothing is right for a case of a day or two. Relaxing only the free troposphere, above 700 hPa, leaves the boundary layer to evolve on its own, which is what we would choose for land-atmosphere work because it is the layer the surface communicates

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with. Relaxing the whole column is the robust choice for longer runs, at the cost of pulling near-surface temperature and humidity toward the reanalysis so that they no longer respond freely to the land surface. That cost is real but bounded: surface fluxes are still diagnosed from the land state, and the soil still evolves on its own, so what the land surface does remains visible even when the air above it is constrained.

The second control fades the advective tendencies and the pressure velocity to zero over a chosen depth above the ground. Two things motivate it. A horizontal gradient taken on a pressure surface lying close to the ground describes the terrain rather than the flow, because the surface intersects the hillsides at a different height at each point of the stencil, and over pronounced relief the resulting spurious tendency does not average away. The pressure velocity must in any case vanish at the ground, which ERA5 does not deliver where the lowest levels sit within the relief. At US-Whs, where the surface varies by 155 m across the stencil, the untapered fields carry a sustained 3 to 5 K per day of low level cooling and a monthly mean pressure velocity near 0.09 Pa s^{-1} of ascent, against 1 K per day and 0.02 Pa s^{-1} at the gentler US-MMS. Both US-Whs and US-MMS are FLUXNET sites that we use as worked examples, and they are described in the example application section below.

We had expected the taper to substitute for nudging at rough sites, and it does not. Tapering alone leaves US-Whs 12 K too cold over a month, because the low level advection there is at once contaminated by terrain and the real driver of the January warming, so removing it discards the signal along with the error. Used together with relaxation of the whole column the taper does help, bringing the mean error at that site from 1.1 to 0.6 K, since the relaxation then has less spurious forcing to work against. We report this because the combination that works is not the one that the physical argument alone would suggest.

Architecture and performance

The workflow is organized as three stages, which are shown schematically in Figure 1. Each stage is exposed as its own subcommand and can be used on its own, so that a user who already holds ERA5 output can enter the workflow at the second stage. A fourth subcommand chains all three together, and the final stage can install the case file and namelist directly into an SCM directory tree so that nothing needs to be copied by hand.

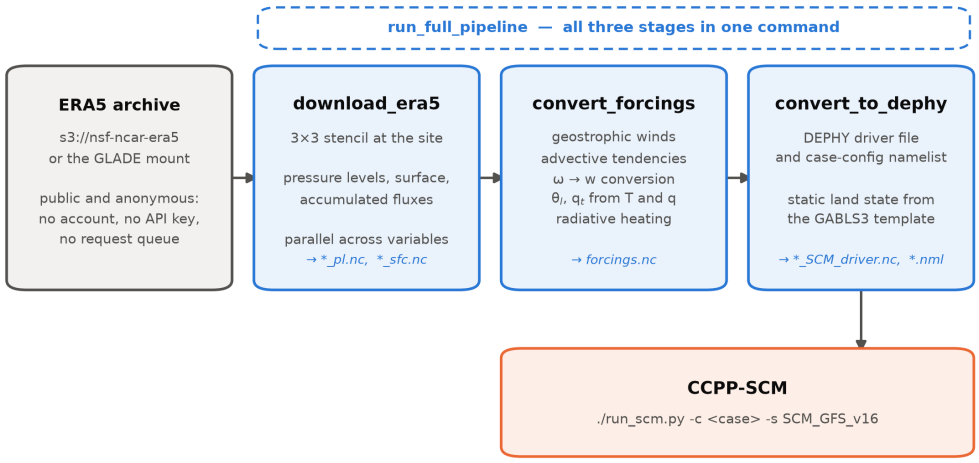


Figure 1: A schematic representation of the era5-to-ccpp-scm-tool software flow.

The default three by three stencil is the smallest footprint that supports centered horizontal gradients, but its width can be increased in whole grid cells, which trades spatial resolution for

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gradients that are less sensitive to noise in the reanalysis fields.

One property of the archive shapes the design of the extraction stage. ERA5 is stored with one whole global field per compressed chunk, so reading a three by three stencil costs the same as reading the entire file, which is roughly 1.3 GB per pressure level variable per day. That cost is incurred per file rather than per site, so we provide a function that extracts many sites in a single pass over the archive, for which the marginal cost of an additional site is close to zero.

Example application

To demonstrate the full workflow we generated cases at two contrasting AmeriFlux sites: US-Whs, a semi-arid shrubland at Walnut Gulch in southern Arizona, beginning 1 January 2019, and US-MMS, a temperate deciduous forest at Morgan Monroe in Indiana, beginning 15 June 2019. Both cases were produced with a single invocation of the full pipeline, contained every variable that the reference DEPHY case file carries, and ran for a month in CCPP-SCM v7.0.0 under the `SCM_GFS_v16` suite without further intervention. Both use relaxation of the whole column and an advection taper of 15000 Pa, which are the settings described above. The relaxation is applied by the SCM at every model time step toward the hourly reanalysis profiles that the case file carries, with the default relaxation timescale of three hours. Figure 2 shows the initial state that was written into each case file along with the resulting simulation. To isolate what the derived land state contributes, we also ran each site a second time with the soil and vegetation state of the stock CCPP-SCM case template in place of the one derived from ERA5, leaving everything else unchanged.

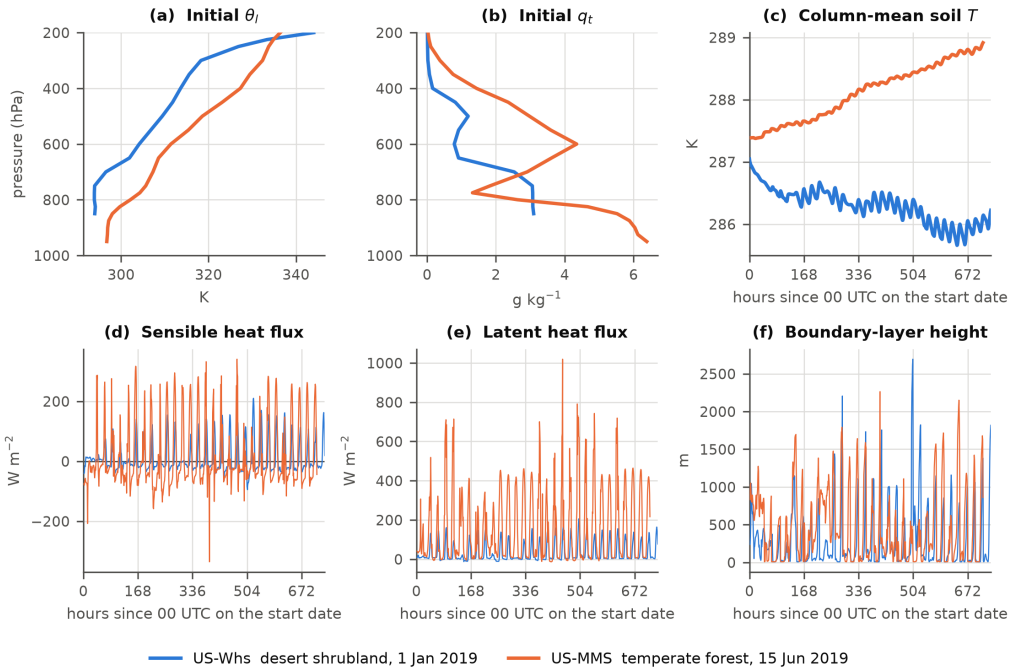


Figure 2: Two cases from generation to simulation. The top row shows the initial state written into the DEPHY case files, and the bottom row shows the month of CCPP-SCM simulation that follows. Panel (c) tracks the column mean soil temperature through the run. The flux panels begin at the third output step, since starting Noah from a surface that HTESSEL equilibrated leaves a transient over the first step. The scripts used to generate this figure are included in the repository.

The atmospheric initial state is specific to the site and season, with US-MMS carrying roughly twice the near-surface total water of US-Whs, at 6.4 against 3.1 g/kg. The land surface state

is likewise specific to each site. The initial soil temperature profiles differ between the two cases in both magnitude and in the sign of their vertical gradient. The January US-Whs profile rises from 280 K in the uppermost layer to 291 K at two meters, as the deep soil retains the previous summer's heat, while the June US-MMS profile falls from 292 K to 285 K over the same depth. Recovering this seasonal reversal follows directly from taking the soil state from the reanalysis, since any single template can supply only one of the two. The two cases also receive different soil textures, silt loam against loam, different land cover classes, grassland against deciduous broadleaf forest, and green vegetation fractions of 0.30 and 0.88.

Both cases track the reanalysis they were built from. Compared against ERA5 2 m temperature over the month, the daily mean air temperature at the lowest model level carries a bias of 0.6 K at US-Whs and 0.2 K at US-MMS, with daily root mean square errors of 1.2 and 1.0 K and correlations of 0.95 and 0.92. We regard this comparison, which the package produces as a diagnostic, as the check to run before trusting a long case, since a column can remain numerically well behaved while drifting away from anything observed.

The simulated partitioning of the surface energy budget follows from the atmospheric and land states together. Taken over daylight hours across the month, the evaporative fraction is 0.80 at the forest site against 0.58 at the desert site, with mean latent heat fluxes of 270 and 76 W/m^2. Building the same two cases on the template soil instead gives evaporative fractions of 0.90 and 0.79 and mean latent heat fluxes of 268 and 114 W/m^2. The template carries a soil moisture saturation fraction of 0.33 at all four levels, which is close to field capacity, together with a vegetation cover of 0.75 that applies all year, and it therefore supplies more water to both surfaces than the reanalysis holds at either. The effect is much the larger at the desert site, where the template raises the evaporative fraction by 0.21 and half again as much latent heat, and it is the site where the template is furthest from the truth. The two sites were run in different seasons, so we present these results as a demonstration that the tool resolves differences between sites and seasons rather than as a controlled attribution of those differences to vegetation type.

The initial land state continues to matter for as long as we ran the cases. The column mean soil temperature of the two US-Whs runs begins 1.8 K apart and ends the month 3.1 K apart, and the two US-MMS runs begin 2.1 K apart and end 1.7 K apart, so in neither case does the difference wash out. The same separation is visible in the air above: the template-initialized runs are persistently 1 to 2 K cooler at both sites through the whole month. Initializing the land surface from the site rather than from a template therefore changes not only the first day of a simulation but the state the column occupies weeks later, which is the clearest statement we can make of what the package is for.

Limitations and future work

Land surface state is taken from the reanalysis, but three caveats attach to the way it gets there. ERA5 describes land cover and soil texture on the same 0.25 degree grid as everything else, which is coarse next to the footprint of an eddy covariance tower, and where a site publishes its own classification we expect that to be used instead. The package accepts a vegetation class by its FLUXNET abbreviation for exactly this purpose, both when a case is built and as a patch applied to a case that has already been written. The HTESSEL and Noah-MP parameter tables also differ in ways that transferring the degree of saturation reduces without removing, so soil moisture carried between them is better read as an equivalent wetness than as a measured water content. Noah-MP moreover begins from a surface that was equilibrated by a different land surface model, and at some sites this produces a transient in the surface fluxes over the first model step, which we recommend discarding and which lengthening the spinup period does not remove. A few fields have no ERA5 counterpart at all, among them the maximum snow albedo, the fractional coverage weights that divide the albedo between its strong and weak zenith angle components, and the friction velocity, and these are still inherited from the case template (Niu et al., 2011). Code modifications are required to use

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this package with Noah-MP configurations that do not use the historical default four layer soil column discretization noted above.

A second group of limitations concerns how long a case can usefully be run, and what has to be given up to run it that long. Left unconstrained the column drifts, so a run of more than about a week needs the relaxation described in the methods, and the setting that proved robust at both of our sites relaxes the whole column rather than the free troposphere alone. The boundary layer is how the land surface communicates with the atmosphere, and relaxing it pulls near-surface temperature and humidity toward the reanalysis, so a study of how strongly the surface controls the air above it may understate the coupling. However, fluxes are still diagnosed from the land state, the soil still evolves freely, and the separation between our two soil initializations persists for the whole month. A user whose question is about the boundary layer rather than about the surface should prefer relaxing only the free troposphere, accept a shorter run, and check the result against the reanalysis before trusting it.

The advection taper carries its own caveat. It exists because horizontal gradients taken close to the ground describe terrain rather than flow, but at a site with pronounced relief that same low level forcing is also carrying the real synoptic signal, and the taper cannot separate the two. Tapering therefore removes real advection along with the spurious part, which is why we use it alongside relaxation rather than in place of it. Neither control is a substitute for the underlying difficulty, which is that a pressure level framework describes the atmosphere near steep terrain poorly. Sites in flat country need neither control for runs of a few days.

Two further caveats follow from the source data. The radiative heating rate that we diagnose is a bulk column mean value, because ERA5 archives radiative fluxes only at the top of the atmosphere and at the surface, and the vertical structure of the heating cannot be recovered from boundary values alone. The cases that we write therefore enable the internal radiation scheme of the SCM, which computes radiation directly and ignores the prescribed rate. Geostrophic winds are also undefined at the equator and unreliable near it, so cases generated in the deep tropics should be treated with caution.

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